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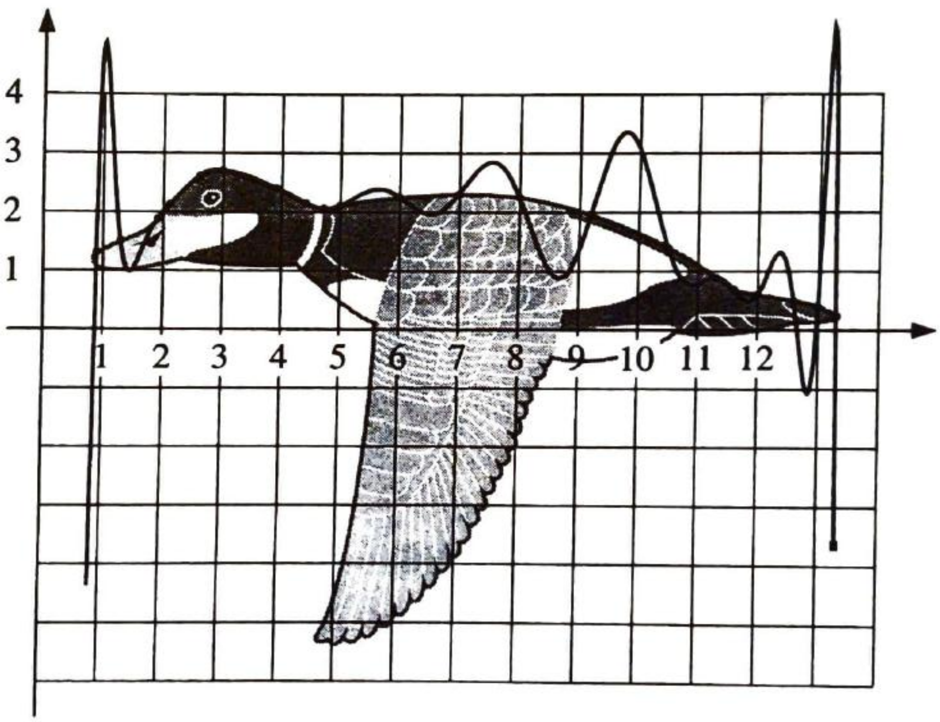

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Math Survey

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From the Editor-in-Chief

January 9, 1995

As editor-in-chief, I would like to thank you for reading Math Survey, Stuyvesant High School's official mathematics publication. Without you, the readers, the buyers and the generous advertisers/contributors, the success of this magazine would not have been possible. My sincerest thanks go out to all my dedicated and hardworking staff members and editors that have contributed to the production of this publication. I would also like to take this opportunity to also thank Mr. Clancy and Dr. Richard Rothenberg for all their advice and counsel.

This school year, Math Survey has decided to do something new: we are publishing one issue of Math Survey earlier to meet the needs of all those A.P. Calculus students that study from our Calculus Review and to satisfy all the math fans that have urged us to publish earlier and more frequently.

This is our first Fall issue in the history of Math Survey, and I hope you enjoy reading it.

Sincerely,

Iris Lan

Iris Lan

15 NEW PIVOTING METHODS WITH 2 NEW SCALES

Iris Lan

ABSTRACT

This paper will introduce 15 new pivoting methods with 2 new scales for getting solutions to linear systems. These 15 new pivoting strategies are developed by manipulating rows and columns and by combining single pivoting methods. New scales factors are defined from the view of the rows and the entire matrix. The accurate and stable solutions can be analyzed by 18 pivoting which include 3 pivoting methods from the paper, "A Geometric Analysis of Gaussian Elimination II" and 15 new pivoting methods.

I. INTRODUCTION

Given a set of n linear simultaneous equations, the general accepted pivoting methods used to solve the system are:

1. Complete Pivoting (CP) 2. Partial Pivoting (PP) 3. Rook's Pivoting (RP)

"A Geometric Analysis of Gaussian Elimination II" by G. Pool and L. Neal used the above pivoting methods obtain answers. This paper will combine the methods to make 12 new pivoting methods. 3 new partial pivoting methods will also be developed.

The following 15 new pivoting methods will be explored in this paper:

i. **3** New Partial Pivoting Methods
1) PP by Full Column **2)** PP by Half Row **3)** PP by Full Row

ii. These **12** pivoting methods combined 3 traditional pivoting methods found in the "A Geometric Analysis of Gaussian Elimination II".

Double Pivoting Methods
4) PP+CP **5)** PP+RP **6)** CP+PP **7)** CP+RP **8)** RP+CP **9)** RP+PP

Triple Pivoting Methods
10) CP+RP+PP **11)** CP+PP+RP **12)** PP+CP+RP
13) PP+RP+CP **14)** RP+PP+CP **15)** RP+CP+PP

Besides pivoting, the scaling factors also influence the solutions. In the paper, "When is Scaling Beneficial ?," only the largest absolute value of each row is used as the scale.

In this paper, I will develop 2 new extra scaling factors:
1. The coefficient average of each row
2. The coefficient average of the entire matrix

A complete example will be given later in order to analyze how to get the accurate answers through the 18 pivoting methods with scales. Then Gaussian Elimination Methods are used to further solve these simultaneous equations.

In order to control error in the computed solution of a linear system, pivoting, scaling, and the condition number will be explored. In this paper, error analysis gives a more complete understanding of above algorithms. The goal of this paper was to find the most stable solutions.

I used a long double precision (80-bit significants) to generate the answers of 3 examples in this paper. Therefore, these answers contain 19 significants decimal digits.

II. PIVOTING STRATEGIES

We now introduce the 15 new pivoting strategies.

i) What are the new single Partial Pivoting methods ?

The definition of Partial Pivoting, PP, in the "A Geometric Analysis of Gaussian Elimination II", is that it is essential that the pivot strategy
"Chooses the largest nonzero entry in the pivot column that is below the diagonal as the pivot element. And can perform a row interchange. "

Based on this idea, we develop the following 3 new Partial Pivoting methods:

1) PP by Full Column
   Select the element in the kth full column that has the largest absolute value; then determine p such that and perform a row interchange of row k and row p to the diagonal.

$$|a_{k,p}| = \underset{1 \le i \le n}{MAX} |a_{k,i}|$$

2) PP by Half Row
   Select the element in the kth row that is above the diagonal and has the largest absolute value; then determine p such that and perform a row interchange of column k and column p to the diagonal.

$$|a_{k,p}| = \underset{k \le i \le n}{MAX} |a_{k,i}|$$

3) PP by Full Row
   Select the element in the kth full row and has the largest absolute value; then determine p such that and perform a row interchange of column k and column p to the diagonal.

$$|a_{k,p}| = \underset{1 \le i \le n}{MAX} |a_{k,i}|$$

The definitions of CP, PP, and RP in the "A Geometric Analysis of Gaussian Elimination II" are the following:

I. CP (Complete Pivoting)
   Choose the largest nonzero entry in the entire matrix as the pivot.
II. PP (Partial Pivoting)
   Choose the largest nonzero entry in the pivot column that is below the diagonal as the pivot element. And, perform a row interchange.
III. RP (Rook's Pivoting)
   Confine the search for the kth pivot to rows k through n.

This paper uses above CP, PP, and RP to make the following double and triple methods.

ii) What are the new <u>double</u> Pivoting methods?

4) PP+CP
   i) Rearrange the entire matrix A and B with the partial pivoting method.
   ii) Use the complete pivoting method to rearrange a new A and a new B.

5) PP+RP
   i) Rearrange entire matrices A and B with the partial pivoting method.
   ii) Rearrange A and B with RP method.

6) CP+PP
   i) Use the complete pivoting method to make new matrices A and B.
   ii) Rearrange entire matrices A and B with the partial pivoting method.

7) CP+RP
   i) Rearrange entire matrices A and B with the complete pivoting method.
   ii) Rearrange A and B with RP method.

8) RP+PP
   i) Rearrange A and B with RP method.
   ii) Rearrange entire matrices A and B with the partial pivoting method.

9) RP+CP
   i) Rearrange A and B with RP method.
   ii) Use the complete pivoting method to make new A and B.

iii). What are the new <u>triple</u> Pivoting methods ?

10) CP+RP+PP
   i) Use the complete pivoting method to make a new A and B.
   ii) Rearrange A and B with RP method.
   iii) Rearrange entire matrices A and B with the partial pivoting method.

11) CP+PP+RP
   i) Use the complete pivoting method to make a new A and B.
   ii) Rearrange entire matrices A and B with partial pivoting method.
   iii) Rearrange A and B with RP method.

12) PP+CP+RP
   i) Rearrange the entire matrices A and B with partial pivoting method.
   ii) Use the complete pivoting method to make new A and B.
   iii) Rearrange A and B with RP method.

13) PP+RP+CP
   i) Rearrange entire matrices A and B with partial pivoting method.
   ii) Rearrange A and B with RP method.
   iii) Use the complete pivoting method to make new A and B.

14) RP+PP+CP
   i) Rearrange A and B with RP method.
   ii) Rearrange the entire matrices A and B with partial pivoting method.
   iii) Use the complete pivoting method to make new A and B.

15) RP+CP+PP
   i) Rearrange A and B with RP pivoting.
   ii) Rearrange entire matrices A and B with complete pivoting method.
   iii) Rearrange entire matrices A and B with partial pivoting method.

There are 18 answers sets in the example 1. Each set was dependent the definition of each pivoting method as described above.

Afterwards, use Gaussian Elimination with LU factorization to get the solutions. The following examples will use all the 18 entire pivoting methods.

EXAMPLE 1:
Find the solutions of the linear system equations be using the above 18 pivoting methods.

The linear system support is A * X = B, where A is an nxn coefficient matrix and X and B are column vectors of the variables and right-hand sides, respectively.

Given matrix A					matrix B
4.00	4800.00	-3200.00	1600.00	-11200.00	-9596.00
-1.00	-1196.00	1700.00	-1200.00	8400.00	8903.00
2.00	2401.00	-1373.00	7600.00	-9800.00	-2570.00
1.00	1199.00	-1024.00	4102.00	-700.00	3478.00
-2.00	-2398.00	2049.00	-4699.00	14357.00	11358.00

Answer: The 18 pivoting methods are:

1. CP (Complete Pivoting) 2. PP (Partial Pivoting) 3. RP (ROOK'S Pivoting) 4. PP FULL COLUMN 5. PP HALF ROW 6. PP FULL ROW 7. PP+CP 8. PP+RP 9. PP+CP 10. CP+RP 11. RP+CP 12. RP+PP 13. CP+RP+PP 14. CP+PP+RP 15. PP+CP+RP 16. PP+RP+CP 17. RP+PP+CP 18. RP+CP+PP

The above 18 solutions can be obtained from a program which combined the definitions of the pivoting methods with Gaussian Elimination factorization. After comparing the 18 answers, we found that the better solutions of the above system come from options 1, 2, 3, 4, 7, 9, 11, 12, 13, 16, 17, and 18. These options result in round-off errors less than 0.15.

PIVOTING	1	2	3	4	5	6	7	8	9
Round Off X_1	0	0.02	0	0.1	0.39	0.7	0	0.31	0.02
Round Off X_2	0	0	0	0	0	0	0	0	0
Round Off X_3	0	0	0	0	0	0	0	0	0
Round Off X_4	0	0	0	0	0	0	0	0	0
Round Off X_5	0	0	0	0	0	0	0	0	0

PIVOTING	10	11	12	13	14	15	16	17	18
Round Off X_1	0.25	0.15	0.15	0.05	0.32	0.25	0.15	0.15	0.15
Round Off X_2	0	0	0	0	0	0	0	0	0
Round Off X_3	0	0	0	0	0	0	0	0	0
Round Off X_4	0	0	0	0	0	0	0	0	0
Round Off X_5	0	0	0	0	0	0	0	0	0

By comparing the above answers, that were found and used by 15 new, additional pivoting methods, one can see the advantages to the 15 new methods. The solutions are much more flexible and accurate than the solutions from the paper, "A Geometric Analysis of Gaussian Elimination II". This example did not use the scale factor.

III. SCALING

The scaling factors produce no round-off error in the system. Therefore, the division that is produced by scaling factors does not alter the orientation of the linear systems. Thus, scaling could lead to a better method of computation for a given linear system.
The scale is defined as "the greatest absolute value of each row"

The 2 new scale factors are defined in this paper as the following:
i) Coefficient average of each row **ii) Coefficient average of the entire matrix**

The above 3 scales can be arranged as following:

0. No Scale
1. Coefficient average of each row

(NEW SCALES)**2. Coefficient average of the entire matrix**
3. The largest absolute value of each row

The definition of the condition number is given by

$$\|A\|_2 \|A^{-1}\|_2$$

where

$$\|A\|_2 = \underset{X \neq 0}{MAX} \frac{|AX|_2}{|X|}$$

EXAMPLE 2:
Compare solutions of a linear system equation through the above 3 scales and find the best solutions. Given

matrix A : matrix B

400.00	4.00	-3200.00	0.00	-11201.00	-13997.00
-1223.00	-1.00	1230.00	-1201.00	8400.00	7205.00
122.00	2.00	-1373.00	7600.00	0.00	6351.00
13.00	0.00	-1024.00	4102.00	-7001.00	-3910.00
-2001.00	-2.00	0.00	-4699.00	15435.00	8733.00

Answer:

SCALE TYPE	NO SCALE	SCALE 1	SCALE 2	SCALE 3
CONDITION #	1.309373e+004	1.429185e+ 04	1.309373e+004	1.160967e+004
Round Off X_1	0	0	0	0
Round Off X_2	1.0e+17	1.69e+13	2.1e+15	1.07e+14
Round Off X_3	2.1e+15	0	5e+17	1.0e+17
Round Off X_4	0	0	0	1.0e+17
Round Off X_5	0	0	0	0

The above table uses pivoting method PP by half row. As result, SCALE 3 has the smallest condition number; so, the round offs are also the smallest. Therefore, the size of the condition number measures how good a solution is.

IV. BETTER SOLUTIONS

How does one get better solutions with the 18 pivoting methods with 4 scales ?
In short, these three equations should be considered in the following order:

1. Which scale is best for the system of equation ?
2. How is important the condition number ?
3. Which one of the 18 pivoting methods is best ?

The following 2 additional examples can be used to help understand all 18 methods:

EXAMPLE 3:
Find the better of the pivoting methods of this linear matrix. The matrix A and matrix B are the same as the example 2.
Answer: For a large N x N matrix, "double" pivoting methods and "triple" pivoting methods can help to get better solutions. The least error was produced from pivoting methods 11,12, 14, and 18.

SCALE type	scale 3	scale 3	scale 1	scale 2
CONDITION #	1.160967e+004	1.160967e+004	1.429185e+ 04	1.309373e+004
PIVOTING options	**14,18**	**11,12**	**17,18**	**16,17,18**
Round Off X_1	1.0e+15	1.0+e15	0	1.60e+16
Round Off X_2	2.2e+13	1.69e+13	2.3e+14	9.25e+13
Round Off X_3	1.0e+16	2.0e+16	1.0e+16	5e+17
Round Off X_4	1.0e+16	1.0e+16	0	0
Round Off X_5	0	0	0	

The better solutions for this matrix are produced by the methods given by options 1, 2, 4, 6, 7 and 9 with "the entire average" scale. The "largest absolute value of each row" scale can get a stable solutions.

From examples 3, we know that 15 new pivoting methods can have more choices for getting better solutions. The solutions of the large NxN matrix that it produced were very good. One can find the most accurate pivoting method under a certain scale by comparing results with the round off errors.

V. CONCLUSIONS:

This paper develops 15 new pivoting methods. This paper has also introduced 2 new scales, each of which influences the condition number of the matrix. For this reason, if we want to a better solution, we should follow the following procedure:

1. First, choose a scale.
2. Then, compute the condition number.
3. Lastly, choose one of the 18 pivoting methods.

Then, we will be able to find which pivoting method gives the most accurate solutions under a certain scale. 15 new pivoting methods with 2 new scales found in this paper are really much more useful than the 3 pivoting methods from the paper, "A Geometric Analysis of Gaussian Elimination II".

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An Introduction to Graph Theory with Algorithms

BY LINDA HONG

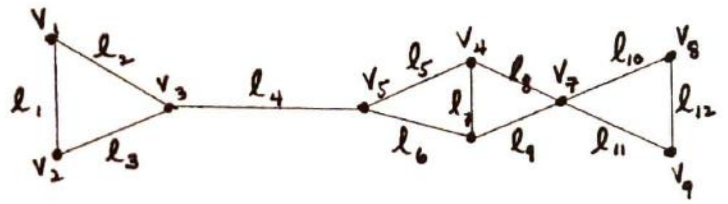
Graph theory is the study of graphs and their properties. Graphs are important for they can be used to solve problems through simple diagrams or complicated algorithm. First, we will define some basic terms commonly found in graph theory.

A graph, $G(V,E)$, consists of a set of vertices, V, and a set of edges, E. Each of the elements of set E can also be defined as an ordered pair of its endpoints which are elements of V. The number of elements in V is denoted as $|V|$ and is called the order of G. Likewise, the number of elements in E is denoted as $|E|$ and is called the size of G.

In a graph a vertex u is said to be adjacent to vertex v if (u,v) is an element of E. An edge (u,v) is defined to be incident with its endpoints u and v. A pair of edges incident with a common vertex is also adjacent. A vertex with no incident edges is said to be isolated. The degree of a vertex is denoted as $d(v)$, and is equal to the number of edges incedent with v. A vertex with an odd degree is called an odd vertex and one with an even degree is called even.

Figure 1 is a graph $G(V,E)$ where $V = \{v_1, v_2, v_3, \ldots, v_9\}$ and $E = \{e_1, e_2, e_3, \ldots, e_{12}\}$. Order $|V| = 9$ and size $|E| = 12$. As v_1 is incident with e_1 and e_2, $d(v_1) = 2$.

Figure 1



Continuing with some more graph terminology, we come to the self-loop, which is an edge (u,v) for which $u = v$. Parallel edges can have the same endpoints, but only in graph theory. However, we will be mainly discussing simple graphs, which have no self-loops or parallel edges.

A graph for which every pair of distinct vertices defines an edge is called a complete graph. A complete graph with n vertices is denoted as K_n. A regular graph is a graph in which all vertices have the same degree measure, k. This is called a k-regular graph. (Note : K_n is also a $(n-1)$-regular). A subgraph of G is obtained by removing a number of vertices or edges. The removal of a vertex also includes the removal of all edges incident to it, whereas the deleting of an edge does not involve removing vertices. If H is a subgraph of G, then G is a supergraph of H and $H \subseteq G$. A subgraph of G induced by a subset of vertices $V' \subseteq V$, consists of all the vertices in V' and the edge of G whose endpoints are both in V'.

A path from a vertex v_1 to v_j in G is an alternating sequence of vertices and edges, $P = v_1, e_1, v_2, e_2, \ldots, e_{j-1}, v_j$, such that for $1 \leq j \leq i$, e_j is incident with v_j and v_{j-1}. It can also be defined simply by a sequence of its vertices and its length in its number of edges path in which each vertex appears only once is a simple path. If $v_1 = v_j$, then P is a cycle or circuit. A simple path with $v_1 = v_j$ is called a simple circuit. Two connected vertices have a path between them, and if every pair of vertices in G is connected, then G is a connected graph. A spanning graph of G is a connected subgraph of G. A cut-edge or cut-vertex which, when removed, disconnects an otherwise connected graph. A trail can be defined as a path allowing repeating vertices. If repeating edges are also allowed, we have a walk.

When a direction is assigned to each edge, we have a directed graph, or more commonly known as a digraph. The order of vertices matters in this case when defining edges as ordered pairs. In edge (u,v), u is its initial vertex or initial endpoint, and v is its terminal vertex or terminal endpoint. u is adjacent *to* v and v is adjacent from u. For any vertex, v, the out-degree is denoted as $d^+(v)$ and the in-degree is $d^-(v)$. A symmetric digraph is a digraph in which for every edge (u,v) there is an edge (v,u). If $d^+(v) = d^-(v)$ for every vertex of a digraph, the digraph is balanced.

Here we also have weighted graphs where each edge is assigned a number, which can also be referred to as the length of the edge. For any edge e in G, $w(e)$ is the weight of the edge. The total weight of the graph is the sum of the weight of its edges.

Now that we have explained some basic terms, we come to our first theorem:

Theorem 1: The number of odd degree vertices in a graph is even.
Proof: By adding up all the degrees of all the vertices of a graph, we will get twice the number of edges because each edge is counted once by each of its two endpoints. Thus,

$$\sum_i d(v_i) = 2|E|$$

The right hand side of this equation must be even and the sum of the even-degree vertices must also be even, and using the rules of addition we can conclude that the sum of the odd-degree vertices is even. Therefore, there must be an even number of odd-degree vertices in any graph.

The complexity of an algorithm is basically the number of steps it takes to transform the input data into the output data. It can also be termed as a function of the quantity of input data, called the problem size. For a problem size s, the complexity of a graph algorithm A is denoted by $C_A(s)$. This is the maximum number of computational stepts required for the execution of A. If algorithm A is faster than algorithm B, A is said to be of lower order than B. In determining the algorithm's overall order, function's low order terms are not relevant. For example, the polynomial $3n^3 + 6n^2 + n + 6$ can be simplified to $O(3n^3)$.

Dijkstra's Algorithm: Find the shortest path from vertex u to vertex v.

For an undirected graph, $G(V,E)$, we replace each edge (u,v) with two directed edges, (u,v) and (v,u). Each vertex v is assigned a label $L(v)$ which will be the length from u. At first $L(v)$ will be assigned the value equal to the weight $w(u,v)$ of the edge (u,v). If $(u,v) \notin E$ then $w(u,v) = \infty$, and $w(v,v) = 0$. We construct set T, a subset of V, so that the current shortest path from u to any $v \notin T$ only passes through vertices in T. The algorithm runs as follows:

1. For all $v \neq u$, $L(v) \leftarrow w(u,v)$
2. $L(u) \leftarrow 0$
3. $T \leftarrow \{u\}$
4. While $T \neq V$ do
   begin
5. find a $v' \notin T$ such that for all $V \notin T$, $L(v') \leq L(v)$
6. $T \leftarrow T \cup \{v'\}$
7. for all $v \notin T$

$$L(v) \leftarrow \min[L(v), L(v') + w(v,v')]$$

   end.

Proof: by induction

(a) for every $v \in T$, $L(v)$ is equal to the length of the shortest path from u to v, and,

(b) for every $v \in T$, $L(v)$ is equal to the length of the shortest path from u to v which, apart from from v, passes only through vertices in T.

from from v, passes only through vertices in T.

When $|T| = 1$ the algorithm satisfies both (a) and (b).

In line 6, v', just before being added to T, is the shortest path from u to v' that utilizes points only in T. To prove algorithm, we will assume that when v' is added to T, $L(v')$ is not the shortest path from u to v', and therefore the shortest path to v' has to pass through at least one vertex not in T, which we will call v''. So, $L(v') < L(v'')$ when v' is added to T. This contradicts line 5 of the program, so in fact there is no shorter path than $L(v')$ when v' is added to T. And so (a) and (b) hold true.

In line 5 there are $O(n)$ comparisons, and from line 4, it is executed $(n-1)$ times. The complexity of this algorithm is $(n)(n-1)$ or simply $O(n^2)$.

Following this comes the longest simple path algorithm which is supposed to find the maximum-length simple path between two specified vertices, u and t, of a graph. This is a fairly simple algorithm with a complexity of $2^{|E|}$ as there are $2^{|E|}$ subsets of E. This algorithm is considered entirely inefficient because of this, though it is quite fun to figure out.

1. MAXP $\leftarrow$ 0
2. for all subsets $E' \subseteq E$ do
3. if E' is a simple path from u to t
   then MAXP $\leftarrow$ max $[w(E'), \text{MAXP}]$

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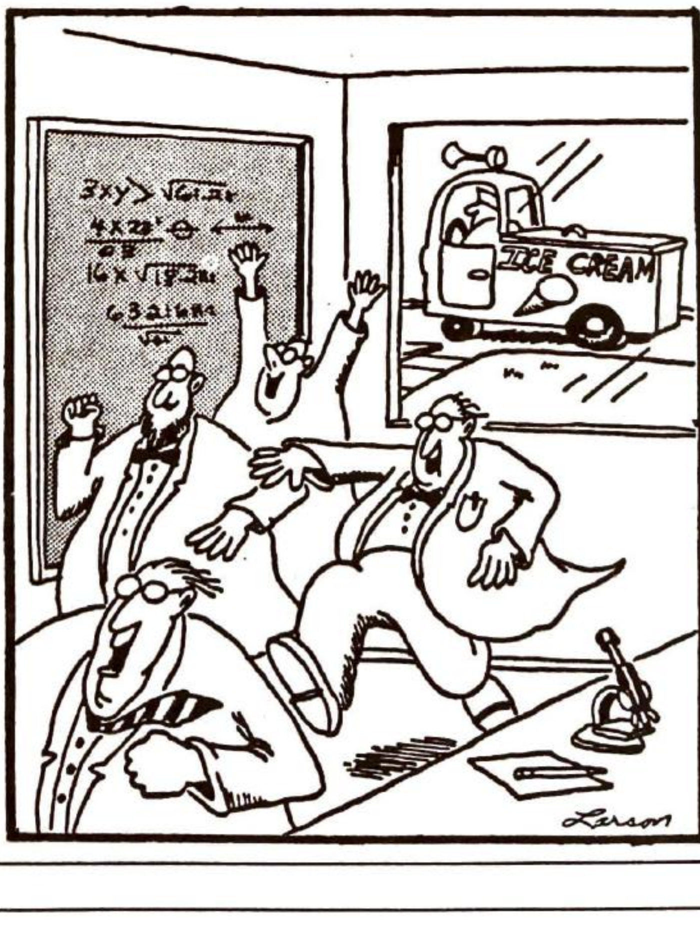

Prime Distribution Theorems

BY JOHNNY CHEN

There are various theorems that give a sense of the distribution of prime numbers such as Bertrand's Postulate, the Riemann zeta function, and the prime number theorem.

Bertrand's Postulate was first posed by the French mathematician Joseph Bertrand in 1845. His claim was that between the positive integers n and $2n$ there exists at least one prime number. Bertrand confirmed this conjecture for all n up to six million. In 1851 the Russian mathematician Chebyshev proved it for all n. An interesting note about the postulate is that one of its results is another proof for the infinitude of primes. As n grows larger, there is always a prime less than $2n$ and greater than n, which means that there are infinitely many primes.

The Riemann zeta-function was not discovered by Riemann. Instead it was Euler who first noticed this relation about a positive integer N, called the Euler product:

$$\prod\left(1-\frac{1}{p^{s}}\right)^{-\phi}=\prod\left(1+p^{-s}+p^{-2s}+\ldots\right)=\sum m \qquad (1)$$

The resulting summation is very similar to the zeta-function, the difference being that the function is the sum of all $1/m^s$ from m=1 to infinity (the exponent s is a complex variable $a+bi$).

The statement in (1) is true because of the formula for the sum of an infinite geometric sequence with the common ratio less than 1 (with r as the common ratio, the sum equals the product of the first term divided by 1-r). In this case, p^{0-s}, or 1, is the first term, so the common ratio is p^{-s}.

The last part of the equation is the sum m to the negative power. m is unique and makes this product distinctive because it runs through all integers that are divisible only by primes less than or equal to N. Therefore the results are all of the possible arrangements of the primes less than or equal to N as factors.

What makes the Riemann zeta-function so important in relation to number theory is the body of work that has emerged from it. Riemann studied the Euler product and modified it. He later found many of its properties, and other mathematicians have used them to prove their own theorems. As a result, the Riemann zeta-function is a central part of the more complex aspects of number theory.

A particular conjecture related to the property of the zeta-function that is now believed to be true, the Riemann hypothesis, has been used to discover many results about the distribution of primes. The prime number theorem was also proved as a result of this function.

The prime number theorem was proved independently by Hadamard and de la Vallee Poussin in 1896. It is stated as follows:

$$\pi(x)\approx\lim_{x\to\infty}\frac{x}{\ln x}$$

This function, $\pi(x)$, represents the number of primes less than or equal to x. This value is asymptotic to $x/\log x$ as x becomes larger -- the actual value and the approximation get closer as they approach infinity. The prime number theorem does not produce an exact result but rather an approximation. This is evident when several values of the function are examined.

The values generated by the prime number theorem are quite close to their actual values. Even so, the theorem could be further refined assuming that the Riemann hypothesis is true. However, this and much of the Riemann zeta-function, as well as related results venture into complex number theory.

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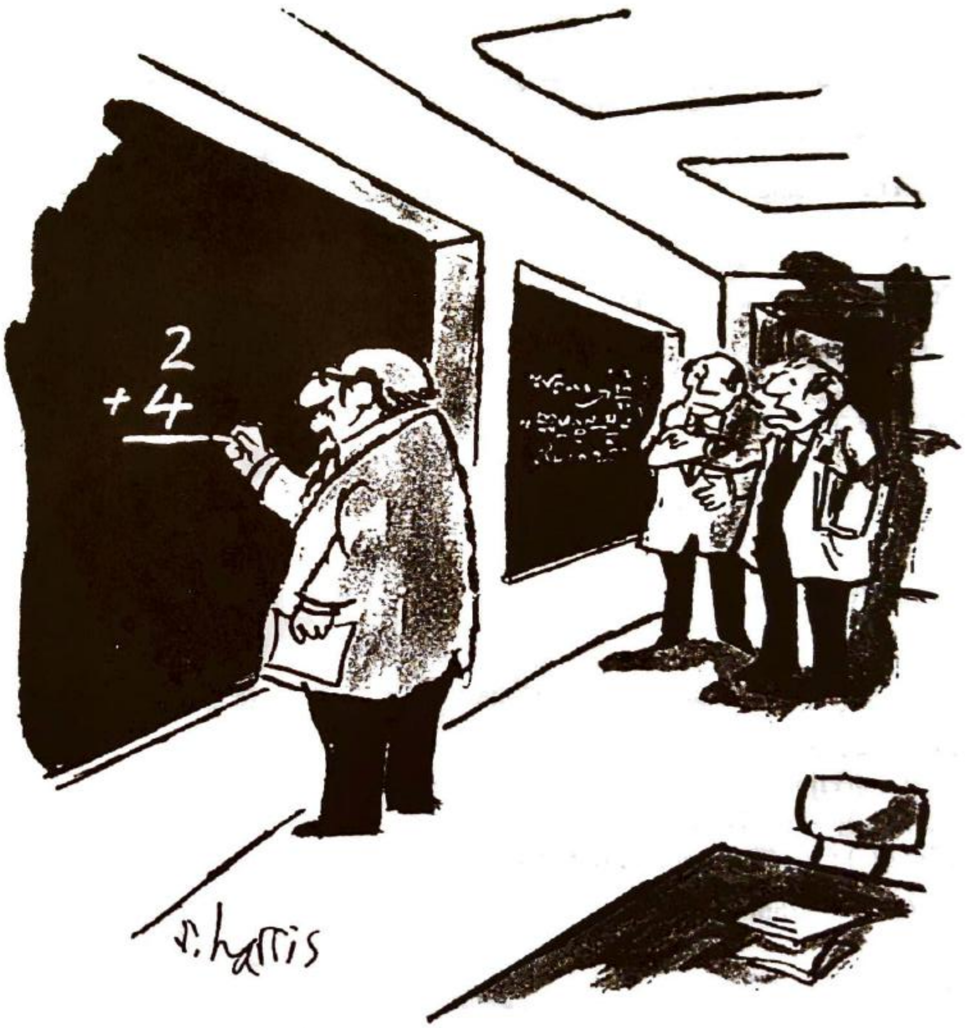


"He was very big in Vienna."

A.P. Calculus Review

Originally by David Z. Weinstein, Updated by *Math Survey* Staff

All material is covered by BC calculus; asterisks (*) indicate material that does not apply to AB calculus.

Limits

1.* Def'n: Let $f(x)$ be defined in some open interval containing a (except possibly at a itself), then $\lim_{x\to a} f(x) = L$, if for any $\varepsilon > 0$ (however small) there exist a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ ($L - \varepsilon < f(x) < L + \varepsilon$) whenever $0 < |x - a| < \delta$ ($a - \delta < x < a + \delta$).

   a.* Def'n: The statement $\lim_{x\to\infty} f(x) = L$ means that for every $\varepsilon > 0$, there exists an $M > 0$ such that $|f(x) - L| < \varepsilon$ whenever $x > M$.

   b.* Def'n: The statement $\lim_{x\to-\infty} f(x) = L$ means that for every $\varepsilon > 0$, there exists an $N < 0$ such that $|f(x) - L| < \varepsilon$ whenever $x < N$.

   c.* Def'n: The statement $\lim_{x\to c} f(x) = \infty$ means that for each $M > 0$, there exists a $\delta > 0$ such that $f(x) > M$ whenever $0 < |x - c| < \delta$.

   d.* Def'n: The statement $\lim_{x\to c} f(x) = -\infty$ means that for each $N < 0$, there exists a $\delta > 0$ such that $f(x) < N$ whenever $0 < |x - c| < \delta$.

2. Limit May Fail to Exist if

   a. $\lim_{x\to a^+} f(x) \neq \lim_{x\to a^-} f(x)$ (e.g. $\lim_{x\to 0} \frac{|x|}{x}$).

   b. values of $f(x) \to \pm\infty$ (e.g. $\lim_{x\to 0} \frac{1}{x}$).

   c. values of $f(x)$ may oscillate indefinitely, approaching no limit (e.g. $\lim_{x\to 0} \sin\frac{1}{x}$).

3. Basic Limit Theorems:
   a. $\lim [f(x) \pm g(x)] = \lim f(x) \pm \lim g(x)$.
   b. $\lim [f(x)\cdot g(x)] = \lim f(x)\cdot \lim g(x)$.
   c. $\lim [f(x)/g(x)] = \lim f(x)/\lim g(x)$ ($\lim g(x) \neq$ to 0).
   d. $\lim [\sqrt[n]{f(x)}] = \sqrt[n]{\lim f(x)}$.

Continuity

1. Def'n: f is continuous at a provided $f(a)$ is defined, $\lim_{x\to a} f(x)$ exists, and $\lim_{x\to a} f(x) = f(a)$ (equivalently: $f(x)$ is continuous at $x = a$, if for any $\varepsilon > 0$, there exists a $\delta > 0$, such that $|f(x) - f(a)| < \varepsilon$ whenever $|x - a| < \delta$.
2. Important Limit Theorems
   a. If f and g are continuous functions, then fg, cf, $f \pm g$, f/g ($g \neq 0$) are also continuous.
   b. All polynomials are continuous.

c. A function continuous on a closed interval $[a, b]$ has the following properties.
   i. $f(x)$ has a minimum and maximum value on the interval.
   ii. $f(x)$ cannot pass from one value to another in the interval without taking every intermediate value (i.e. It's an unbroken line).

Derivatives

1. Def'n: let $x = a$ be on the curve $y = f(x)$. At each such point where $\lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$ exists, the function is said to be differentiable (have a derivative). The limit is called the derivative of y with respect to x and denoted by $\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$. Additionally, $f'(a) = \lim_{\Delta x \to 0} \frac{f(a+\Delta x) - f(a)}{\Delta x} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ (derivative at a). If $y = f(x)$ then
   a. $dy/dx = f'(x) = D_x y = y'$ (first derivative).
   b. $d^2y/dx^2 = f''(x) = D^2_x y = y''$ (second derivative).
   c. $d^3y/dx^3 = f'''(x) = D^3_x y = y'''$ (third derivative).
2. Formulas for Derivatives
   c, n are constants; u, v are differentiable functions of x.

1. $$\frac{d(c)}{dx} = 0$$

2. $$\frac{d(cu)}{dx} = c\frac{du}{dx}$$

3. $$\frac{d(u^n)}{dx} = nu^{n-1}\frac{du}{dx}$$

4. $$\frac{d(u+v)}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

5. $$\frac{d(uv)}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

6. $$\frac{d\left(\frac{u}{v}\right)}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

7. $$\frac{d(\sin u)}{dx} = \cos u\frac{du}{dx}$$

8. $$\frac{d(\cos u)}{dx} = -\sin u\frac{du}{dx}$$

9. $$\frac{d(\tan u)}{dx} = \sec^2 u\frac{du}{dx}$$

10. $$\frac{d(\cot u)}{dx} = -\csc^2 u\frac{du}{dx}$$

11. $$\frac{d(\sec u)}{dx} = \sec u \tan u\frac{du}{dx}$$

12. $$\frac{d(\csc u)}{dx} = -\csc u \cot u\frac{du}{dx}$$

13. $$\frac{d(\ln u)}{dx} = \frac{1}{u}\frac{du}{dx}$$

14. $$\frac{d(e^u)}{dx} = e^u\frac{du}{dx}$$

15. $$\frac{d(\log_c u)}{dx} = \log_c e \cdot \frac{1}{u}\frac{du}{dx}$$

16. $$\frac{d(a^u)}{dx} = a^u \ln a\frac{du}{dx}$$

17. $$\frac{d(\sin^{-1} u)}{dx} = \frac{1}{\sqrt{1-u^2}}\frac{du}{dx}$$

18. $$\frac{d(\cos^{-1} u)}{dx} = -\frac{1}{\sqrt{1-u^2}}\frac{du}{dx}$$

19. $$\frac{d(\tan^{-1} u)}{dx} = \frac{1}{1+u^2}\frac{du}{dx}$$

20. $$\frac{d(\cot^{-1} u)}{dx} = -\frac{1}{1+u^2}\frac{du}{dx}$$

21. $$\frac{d(\sec^{-1} u)}{dx} = \frac{1}{\sqrt{u^2(u^2-1)}}\frac{du}{dx}$$

22. $$\frac{d(\csc^{-1} u)}{dx} = -\frac{1}{\sqrt{u^2(u^2-1)}}\frac{du}{dx}$$

23. Chain Rule (for composite functions)

If $y = f(u)$ and $u = g(x)$ $(y = f(g(x)))$, then $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = f'(u)g'(x) = f'(g(x))g'(x)$.

24. If $y = f(x)$ is single-valued and has nonzero derivatives on a given interval, then the inverse function exists and has a derivative. Suppose $y = f(x)$ has an inverse function (i.e. $x = f^{-1}(y)$ exists) that is differentiable, then $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$.

25. *Parametric Equations

If $y = f(t)$ and $x = g(t)$, then

a.* $y' = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$.

b.* $y'' = \frac{d^2y}{dx^2} = \frac{dy'}{dx} = \frac{\frac{dy'}{dt}}{\frac{dx}{dt}}$.

c.* $y''' = \frac{d^3y}{dx^3} = \frac{dy''}{dx} = \frac{\frac{dy''}{dt}}{\frac{dx}{dt}}$.

26. Logarithmic Differentiation

If $y = x^{g(x)}$ then $\ln y = \ln x^{g(x)} = g(x) \ln x$. Differentiating yields $\frac{1}{y}\frac{dy}{dx} = g'(x)\ln x + g(x)\frac{1}{x}$ or $\frac{dy}{dx} = y\left(g'(x)\ln x + g(x)\frac{1}{x}\right)$.

Curve Sketching

1. Without Derivatives
   a. Intercepts: To get the x intercepts, set $y = 0$ and solve for x; to get the y intercepts, set $x = 0$ and solve for y.
   b. Symmetry: If a curve has form $y = F(x)$ or $f(x, y) = 0$, then it is
      i. Symmetric to x-axis if $f(x, y) = f(x, -y)$ (or $F(x) = -F(x)$).
      ii. Symmetric to y-axis if $f(x, y) = f(-x, y)$ (or $F(x) = F(-x)$).
      iii. Symmetric to origin if $f(x, y) = f(-x, -y)$ (or $F(x) = -F(x)$).
   c. Extent (domain and range): Find x and y values on which the function is defined.
   d. Asymptotes:
      i. Horizontal Asymptote: Let $x \to \pm\infty$ and find the limiting value(s) of y.
      ii. Vertical Asymptote: Let $y \to \pm\infty$ and find the limiting value(s) of x.
      iii. Oblique Asymptote $y = mx + b$ $(m \neq 0)$ is an asymptote of $y = f(x)$ if $\lim_{x\to\infty}[f(x) - (mx + b)] = 0$.
      iv. We can also solve for one variable in terms of the other; set denominator equal to 0, etc. For

example, if $y = f(x) = g(x) + \dfrac{c}{h(x)}$, then $y = g(x)$ is a curved (e.g. parabolic, oblique, or horizontal) asymptote, and $h(x) = 0$ represents any vertical asymptotes.

2. With Derivatives
   a. Def'n:
      i. A function $f(x)$ is increasing (decreasing) on an interval if whenever x_1 and x_2 belong to the interval and $x_2 > x_1$, then either $f(x_2) > f(x_1)$ (rising curve) or $f(x_2) < f(x_1)$ (falling curve).
      ii. $f(x)$ is a smooth function if both $f(x)$ and $f'(x)$ are continuous.
      iii. Relative Extremum: maximum or minimum value.
      iv. Critical values of $f(x)$ are values where $f'(x) = 0$ or is undefined.
      v. A smooth curve is concave upward at a point if successive tangents to the curve from that point turn counterclockwise (slope increases - curve lies above tangent). A smooth curve is concave downward at a point if successive tangents to the curve from that point turn clockwise (slope decreases - curve lies below tangent).

      vi. Point of inflection: a point where the curve changes its concavity.

   b. Important Theorems:
      i. Function $f(x)$ is increasing at $x = a$ if $f'(a) > 0$, decreasing at $x = a$ if $f'(a) < 0$.
      ii. If $f(x)$, a continuous, differentiable function, has a relative extrema at $x = a$, then $f'(a) = 0$ (but not conversely).

iii. Tests for Maxima and Minima

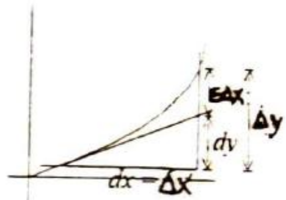
(1) Max: $f'(a) = 0$ and $f'(x)$ changes from + to - at $x = a$.
    Min: $f'(a) = 0$ and $f'(x)$ changes from - to + at $x = a$.
(2) Max: $f'(a) = 0$ and $f''(a) < 0$.
    Min: $f'(a) = 0$ and $f''(a) > 0$.

      iv. Curve is concave up at $x = a$ if $f''(a) > 0$; concave down if $f''(a) < 0$. (If $f''(a) = 0$, it's concave up or down depending on its concavity in the neighborhood of $x = a$.)
      v. If $f(x)$ is a function with continuous second derivative, $x = a$ is a point of inflection of $f(x)$, if $f''(a) = 0$ and $f(a) \neq 0$ or $f''(x)$ changes sign through a.
   c. Absolute extrema can be attained
      i. Where $f'(x) = 0$.
      ii. At the endpoints of interval.
      iii. Where the derivative does not exist.

Applications of the Derivative

1. If a function is differentiable at a point, then it must be continous at the point, but not conversely, i.e.: differentiability implies continuity.
2. Rolle's theorem: If $f(x)$ is continuous on $[a, b]$, $f'(x)$ exists on (a, b), and $f(a) = (b) = 0$, then there exists a point c $(a < c < b)$ such that $f'(c) = 0$.
3. Mean value theorem for derivatives: If $f(x)$ is continuous on $[a, b]$ and $f'(x)$ exists on (a, b) then there exists a point c such that $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.

4. L'Hôpital's Rule
   a. If $f(x)$ and $g(x)$ are in an interval including $x = a$ and satisfy the hypotheses of the general mean value theorem and $\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$, then $\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}$
   b. Evaluation of other indeterminate forms:
      i. $0 \cdot \infty$. Change to $(1/\infty)(\infty)$, which equals ∞/∞.
      ii. $\infty - \infty$. Transform by algebraic or trigonometric manipulation to acceptable form.
      iii.* 1^∞, 0^0, and ∞^0. Take the log of the function and change to one of the above forms.
5. Differentails. Def'n: When $y = f(x)$, the differential of x (independent value) $dx = \Delta x$ and the differential of y (dependent value) $dy = f'(x)\,dx$.
   Motivation for definition:
   a. One may now think of $f'(x)$ as a quotient.
   b. For small Δx, $dy = \Delta y$.



   Note: $f'(x) = \lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x}$, therefore $\frac{\Delta y}{\Delta x} = f'(x) + \varepsilon$ ($\varepsilon \to 0$ when $\Delta x \to 0$) implies $\Delta y = f'(x)\Delta x + \varepsilon\Delta x$. Main part of Δy is $f'(x)\Delta x$ since $\varepsilon\Delta x$ is relatively small. So we define $dy = f'(x)\Delta x$, therefore $\Delta y = dy + \varepsilon\Delta x \approx dy$.
6. Related rates problems: Find the relationship between variables, take the derivative with respect to time, and substitute.
7. Newton's Method for approximating zeros of functions: Let $f(c) = 0$, where f is differentiable on an open interval containing c. Then, to approximate c, we use the following steps:
   a. Make an initial estimate x_1, that is close to c. (A graph is helpful.)
   b. Determine a new approximation by $x_{n+1} = x_n - f(x_n) / f'(x_n)$.
   c. If $|x_n - x_{n+1}|$ is less than the desired accuracy, let x_{n+1} serve as the final approximation. Otherwise, return to step b and calculate a new approximation.

Motion of Particles

1. Rectilinear motion (motion along a straight line)
   If $s = f(t)$, then $v_{Av} = \Delta s/\Delta t$ = (total distance traveled) / (time elapsed) and $a_{Av} = \Delta v/\Delta t$.
   a. $v = ds/dt = f'(t)$. If $v > 0$, then the particle is moving in the direction of increasing s. If $v < 0$, then the particle is moving in direction of decreasing s. If $v = 0$, then the particle is instanteously at rest. At the instant the particle reverses direction, v must be 0, but not conversely.
   b. $a = dv/dt = f''(t)$. The particle changes direction if $v = 0$ and $a \neq 0$ or v changes sign as we go through $v = 0$. The speed of the particle, $|v|$ (magnitude of v), increases when $a > 0$ and $v > 0$ or $a < 0$ and $v < 0$. The speed decreases when $a < 0$ and $v > 0$ or $a > 0$ and $v < 0$.
2.* Curvilinear motion (motion along a curve)

   Parametric equations of motion: $x = f(t)$, $y = g(t)$; $v_x = \frac{dx}{dt}$, $v_y = \frac{dy}{dt}$, $a_x = \frac{dv_x}{dt}$, $a_y = \frac{dv_y}{dt}$.

   $\|\mathbf{v}\| = \sqrt{v_x^{\,2} + v_y^{\,2}}$ (magnitude), $\tan\theta = \frac{v_y}{v_x}$ (quadrant depends on signs of v_x and v_y).

   $\|\mathbf{a}\| = \sqrt{a_x^{\,2} + a_y^{\,2}}$ (magnitude), $\tan\alpha = \frac{a_y}{a_x}$ (quadrant depends on signs of a_x and a_y).

Since $\tan\theta = \frac{\frac{dx}{dt}}{\frac{dy}{dt}} = \frac{dy}{dx}$ and $v^2 = v_x^{\,2} + v_y^{\,2} = \left(\frac{ds}{dt}\right)^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$, $\|\mathbf{v}\| = \left|\frac{ds}{dt}\right|$.

It follows that the velocity vector has is tangent to the curve and the speed of the particle = $\left|\frac{ds}{dt}\right|$, where s is the distance along the curve from some fixed point.

Integration

1. Def'n: Indefinite Integral (Antiderivative) $\int f(x)\,dx = F(x) + C$ if $F'(x) = f(x)$.
   Fundamental Rules:
   a. $\int (u + v)\,dx = \int u\,dx + \int v\,dx$.
   b. $\int ku\,dx = k\int u\,dx$.
   c. $\int u^n\,du = u^{n+1}/(n+1) + C \quad (n \neq -1)$.
2. Def'n: Definite Integral
   Properties:
   a. $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$.
   b. $\int_a^a f(x)\,dx = 0$.
   c. $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$ (for any c).
   d. $\int_a^b k\,dx = k(b - a)$.
   e. $\int_a^b (u + v)\,dx = \int_a^b u\,dx + \int_a^b v\,dx$.
3. Fundamental Theorem of Integral Calculus (Connects concept of indefinite and definite integrals)
   If $f(x)$ is continuous on (a, b), $\int f(x)\,dx = F(x)$ for any particular integral, then

   $\int_a^b f(x)\,dx = F(b) - F(a)$.

Integration Theorems

1. If $f(x)$ is continuous on $[a, b]$, then it is integrable on $[a, b]$. (This works even for finite numbers of discontinuities.)
2. If f and g are integrable on $[a, b]$ and $f(x) \le g(x)$ for all x on $[a, b]$, then $\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$.
3. If f is integrable on $[a, b]$ and $m \le f(x) \le M$ for all x in $[a, b]$, then $m(b - a) \le \int_a^b f(x)\,dx \le M(b - a)$.
4. If $F(x)$ and $G(x)$ are antiderivatives of the same function (i.e. $dF/dx = dG/dx$), then $F(x) = G(x) + C$.
5. Average Value of a Function (with respect to x on $[a, b]$): $y_{av} = \frac{1}{b-a}\int_a^b f(x)\,dx$.

6. Let f be continuous on $[a, b]$. Let x be a number on (a, b) and define $F(x) = \int_a^x f(t)\,dt$, then F is differentiable and $F'(x) = f(x)$.

7. Change on Variable in Define Integrals: $x = \varphi(t)$, $a = \varphi(t_1)$, $b = \varphi(t_2)$, then $\int_a^b f(x)\,dx =$

$$\int_{t_1}^{t_2} f(\varphi(t))\cdot\varphi'(t)\,dt.$$

Approximations of Integrals

1. Trapezoidal Rule: $\int_a^b y\,dx \approx \frac{h}{2}(y_0 + 2y_1 + 2y_2 + \cdots + 2y_{n-1} + y_n)$, $h = \frac{b-a}{n}$.

$$\begin{array}{ll} x_0 = a & y_0 = f(x_0) = f(a) \\ x_1 = a + h & y_1 = f(x_1) \\ x_2 = a + 2h & y_2 = f(x_2) \\ \vdots & \vdots \\ x_n = a + nh & y_n = f(x_n) = f(b) \end{array}$$

$$\varepsilon_T = \frac{-(b-a)h^2 f''(c)}{12}, \ a \le c \le b.$$

2.* Area under parabolic arc: $A = (h/3)(y_0 + 4y_1 + y_2)$, where y_0, y_1, and y_2 are three equidistant ordinates, and $h = y_1 - y_0 = y_2 - y_1$.

3.* Simpson's Rule: $y = f(x) \ge 0$ on $[a, b]$.

$$\int_a^b y\,dx \approx \frac{h}{3}(y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + \cdots + 4y_{n-1} + y_n),\ h = \frac{b-a}{n},\ n \text{ must be even.}$$

$$\varepsilon_s = \frac{-(b-a)h^4 f^{(4)}(c)}{180}, \ a \le c \le b..$$

Basic Integration Formulas

1. $\int c\,f(x)\,dx = c\int f(x)\,dx$.
2. $\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$.
3. $\int u^n\,du = u^{n+1}/(n+1) + C \ (n \ne 1)$.
4. $\int (1/u)\,du = \ln|u| + C$.
5. $\int e^u\,du = e^u + C$.
6. $\int a^u\,du = a^u/\ln a + C \ (a > 0, a \ne 1)$.
7. $\int \cos u\,du = \sin u + C$.
8. $\int \sin u\,du = -\cos u + C$.
9. $\int \tan u\,du = \ln|\sec u| + C = -\ln|\cos u| + C$.
10. $\int \cot u\,du = \ln|\sin u| + C = -\ln|\csc u| + C$.
11. $\int \sec^2 u\,du = \tan u + C$.
12. $\int \csc^2 u\,du = -\cot u + C$.
13. $\int \sec u \tan u\,du = \sec u + C$.
14. $\int \csc u \cot u\,du = -\csc u + C$.
15. $\int \sec u\,du = \ln|\sec u + \tan u| + C$.
16. $\int \csc u\,du = \ln|\csc u - \cot u| + C$.
17. $\int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin\frac{u}{a} + C$.
18. $\int \frac{du}{a^2 + u^2} = \frac{1}{a}\arctan\frac{u}{a} + C$.
19. $\int \frac{du}{|u|\sqrt{a^2 - u^2}} = \frac{1}{a}\text{arcsec}\frac{u}{a} + C$.

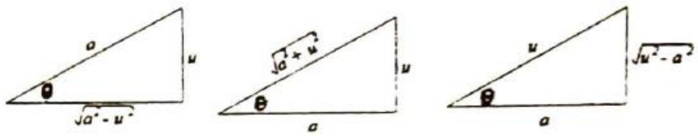
Techniques of Integration

1. Substitution: Use of identities and change of variable.
2. Integration by Parts: $\int u\,dv = uv - \int v\,du$.(From integrating the product rule: $\int d(uv) = \int u\,dv + \int v\,du$.)

   Also Note: For definite integrals: $\int_a^b u\,dv = uv\Big]_a^b - \int_a^b v\,du$. Can be used for: $\int x\,e^x\,dx$, twice for $\int x^2 \cos x\,dx$, twice for $\int e^x \sin x\,dx$, $\int \ln x\,dx$, $\int \arcsin x\,dx$.

3.* Trigonometric Substitution:

   $u = a\sin\theta$ in $\sqrt{a^2 - u^2}$ yields $a\cos\theta$.
   $u = a\tan\theta$ in $\sqrt{a^2 + u^2}$ yields $a\sec\theta$.
   $u = a\sec\theta$ in $\sqrt{u^2 - a^2}$ yields $a\tan\theta$.



4.* Partial Fractions: If the original denominator has

   a.* A linear factor, place $ax + b$ in the new denominator, with a constant numerator: $\frac{A}{ax+b}$.

   b.* A quadratic factor, place $ax^2 + bx + c$ in the new denominator, with a linear numerator: $\frac{Ax+B}{ax^2+bx+c}$.

   c.* An unfactorable polynomial to the r power, place:

$$\frac{A_{n-1}x^{n-1} + A_{n-2}x^{n-2} + \cdots + A_0x^0}{\left(a_nx^n + a_{n-1}x^{n-1} + \cdots + a_0x^0\right)^r} + \frac{B_{n-1}x^{n-1} + B_{n-2}x^{n-2} + \cdots + B_0x^0}{\left(a_nx^n + a_{n-1}x^{n-1} + \cdots + a_0x^0\right)^{r-1}} + \cdots$$

   (until the denominator is raised to the first power).

   d.* Any combinations/repetitions of the above cases should be repeated.

5. For $\int \frac{p(x)}{q(x)}$, divide the numerator by the denominator if the former is of higher or of equal degree to the latter.
6. Completing the Square: Use this technique for quadratic functions if necessary.
7. Splitting up Fractions: $\int \frac{Ax+B}{ax^2+bx+c} = \int \frac{Ax}{ax^2+bx+c} + \int \frac{B}{ax^2+bx+c}$

Applications Of The Definite Integral

1. Areas:

   a. Area bounded by $y = f(x)$, x-axis, lines $x = a$ and $x = b$ $(b > a)$, $A = \int_a^b f(x)\,dx = \int_a^b y\,dx$.

   b. Area bounded by $x = g(y)$, y-axis, lines $y = a$ and $y = b$ $(b > a)$, $A = \int_a^b g(y)\,dy = \int_a^b x\,dy$.

   c. Using area of a circle $= \pi r^2$, we find $\int_0^a \sqrt{a^2 - x^2}\,dx = \frac{1}{4}\pi a^2$.

   d. If $f(x) \geq g(x)$ on $[a, b]$, then the area bounded by the curves $y_1 = f(x)$, $y_2 = g(x)$ and the lines $x = a$

and $x = b$, $A = \int_a^b (y_2 - y_1)\,dx = \int_a^b (g(x) - f(x))\,dx$.

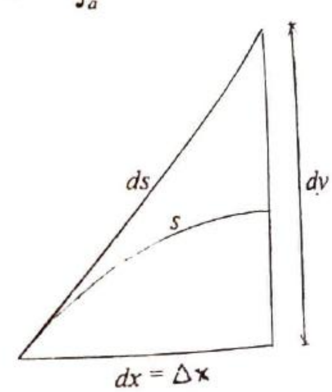
e.* Polar Coordinates:

i.* The area bounded by $r = f(\theta)$ and two radii vectors $\theta = \alpha$ and $\theta = \beta$ is $A = \frac{1}{2}\int_\alpha^\beta r^2\,d\theta$.

ii.* The area between $r_1 = f(\theta)$ and $r_2 = g(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is $A = \frac{1}{2}\int_\alpha^\beta (r_1^2 - r_2^2)\,d\theta$.

2. Volumes:
   a. Known Cross-Sectional Area: Volume of solid with cross-sectional area $A(x)$ perpendicular to x-axis at x is: $V = \lim_{n\to\infty} \sum_{k=1}^{n} A(c_k)\Delta x_k = \int_a^b A(x)\,dx$.
   b. Solid of Revolution: A solid generated by revolving a planar region about an axis in its plane (plane perpendicular to axis cuts circular cross sections).
   c. Volume of solid formed by revolving region bounded by $y = f(x)$, $x = a$, $x = b$ $(b > a)$, about x-axis is $V = \pi\int_a^b y^2\,dx$ (Cylindrical Disks).
   d. Volume of solid formed by revolving region bounded by $x = g(y)$, $y = a$, $y = b$ $(b > a)$, about y-axis is $V = \pi\int_a^b x^2\,dy$.
   e. If $f(x) \geq g(x)$ on $[a, b]$, volume generated by revolving about x-axis region bounded by curves $y_1 = f(x)$, $y_2 = g(x)$, $x = a$ and $x = b$ $(b > a)$ is $V = \pi\int_a^b (y_2^2 - y_1^2)\,dx$ (cylindrical washers).
   f. Volume of solid generated by revolving region bounded by $y = f(x)$, x-axis, $x = a$, $x = b$ $(b > a)$ about y-axis is $V = 2\pi\int_a^b xy\,dx$ (cylindrical shells). Volume of solid generated by revolving region bounded by $x = g(y)$, y-axis, $y = a$, $y = b$ about x-axis $V = 2\pi\int_a^b yx\,dy$.

3.* Arc Length:
   a.* If $y = f(x)$, and $y' = f'(x)$ are continuous, then the arc length of $y = f(x)$ from $x = a$ to $x = b$ $(b > a)$ is $s = \int_a^b \sqrt{1 + y'^2}\,dx$. Similarly for $x = g(y)$, $x' = g'(y)$ from $y = a$ to $y = b$, $s = \int_a^b \sqrt{1 + x'^2}\,dy$.
   b.* The differential of arc for a smooth curve $y = f(x)$, $ds = \sqrt{1 + y'^2}\,dx$. Squaring, we get $(ds)^2 = (dx)^2 + (dy)^2$ and the above formula can be written as $s = \int ds$. Also note the geometric interpretation.
   c.* Parametrically, if $x = f(t)$, $y = g(t)$, then since $(ds)^2 = (dx)^2 + (dy)^2$, taking the square root and integrating yields $s = \int_{t_0}^{t_1} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$.



d.* Polar Coordinates: Length of curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is $s = \int_{\alpha}^{\beta} \sqrt{r^2 + r'^2}\, d\theta$.

4.* Work: Motion Along Straight Line

a.* $W = fd$ (for a constant force only).

b.* Total work done in moving an object along x-axis from a to b, if $f(x)$ is a force exerted at a point x, is $W = \int_a^b f(x)\, dx$.

c.* Hooke's Law: $f \propto x$ or $f = kx$. (k is the spring constant, i.e. the force required to stretch the spring one unit from equilibrium position.)

5.* Improper Integrals

a.* Convergent: approaches a limiting value. Divergent: Approaches no value (∞, oscillates, etc.).

b.* Type 1 (Discontinuous Integrand)

i.* Integrand becomes ∞ at lower limit: $\int_a^b f(x)\, dx = \lim_{h\to 0^+} \int_{a-h}^b f(x)\, dx$.

ii.* Integrand becomes ∞ at upper limit: $\int_a^b f(x)\, dx = \lim_{h\to 0^+} \int_a^{b-h} f(x)\, dx$.

iii.* c ($a < c < b$) makes integrand ∞: $\int_a^b f(x)\, dx = \int_a^c f(x)\, dx + \int_c^b f(x)\, dx$, use i and ii.

c.* Type 2 (Infinite Limits)

i.* $\int_a^{\infty} f(x)\, dx = \lim_{b\to\infty} \int_a^b f(x)\, dx$.

ii.* $\int_{-\infty}^b f(x)dx = \lim_{a\to-\infty} \int_a^b f(x)\, dx$.

iii.* $\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^c f(x)dx + \int_c^{\infty} f(x)\, dx$ (for suitable c).

More on Limits

1. Squeeze Theorem: If $A \le f(x) \le g(x)$ for some constant A and $\lim_{x\to c} g(x) = A$, then $\lim_{x\to c} f(x) = A$.
2. Important Limits:

a. $\lim_{x\to 0} \frac{\sin x}{x} = 1$.

b. $\lim_{x\to 0} \frac{\tan x}{x} = 1$.

c. $\lim_{x\to 0} \frac{1-\cos x}{x} = 0$.

d. $\lim_{x\to 0} \frac{e^x - 1}{x} = 1$.

e. $\lim_{h\to\infty} \left(1 + \frac{1}{h}\right)^h = \lim_{h\to 0} (1+h)^{\frac{1}{h}} = e$.

f. $\lim_{k\to\infty} \left(1 + \frac{n}{k}\right)^k = \lim_{\substack{k\to\infty \\ \left(\frac{k}{n}\to\infty\right)}} \left[\left(1 + \frac{1}{\frac{k}{n}}\right)^{\frac{k}{n}}\right]^n = e^n$

* Note: The rest of the material applies only to the BC curriculum.*

Sequences and Series

1. Definitions
   a. Def'n: A sequence is a function whose domain is the set of positive integers. (The numbers in the range of the sequence are its elements or terms.)
   b. Def'n: A sequence $\{a_n\}$ has the limit L, written $\lim_{n\to\infty} a_n = L$, if to each number $\varepsilon > 0$, there corresponds an integer $N > 0$ such that $|a_n - L| < \varepsilon$ whenever $n \geq N$ (i.e. for any $\varepsilon > 0$ there exists an integer $N > 0$ such that all terms beyond the Nth in the sequence differ from L by less than ε).
   c. Def'n: A sequence that has a limit is said to be convergent, and to converge to that limit. A sequence that fails to have a limit is divergent.
2. Basic Theorems
   a. Thm: If $\lim_{x\to\infty} f(x) = L$ and f is defined for all positive integers, then also $\lim_{n\to\infty} f(n) = L$, where n is a positive integer.
   b. Thm: The limit of a convergent sequence is unique.
   c. Thm: If $\{a_n\}$and $\{b_n\}$ are convergent sequences and C is a constant, then
      i. $\lim_{n\to\infty} ca_n = c \lim_{n\to\infty} a_n$.
      ii. $\lim_{n\to\infty} (a_n \pm b_n) = \lim_{n\to\infty} a_n \pm \lim_{n\to\infty} b_n$.
      iii. $\lim_{n\to\infty} a_n b_n = \lim_{n\to\infty} a_n \lim_{n\to\infty} b_n$.
      iv. $\lim_{n\to\infty} (a_n/b_n) = \lim_{n\to\infty} a_n / \lim_{n\to\infty} b_n$ (provided $\lim_{n\to\infty} b_n \neq 0$).
3. Monotonic Sequences
   a. Def'n: A sequence is said to be monotonic and
      i. Increasing (or non-decreasing) if $a_n \leq a_{n+1}$ (i.e. $a_1 \leq a_2 \leq a_3 \leq \ldots$).
      ii. Strictly increasing if $a_n < a_{n+1}$ (i.e. $a_1 < a_2 < a_3 < \ldots$).
      iii. Decreasing (or non-increasing) if $a_n \geq a_{n+1}$ (i.e. $a_1 \geq a_2 \geq a_3 \geq \ldots$).
      iv. Strictly decreasing if $a_n > a_{n+1}$ (i.e. $a_1 > a_2 > a_3 > \ldots$).
4. Bounds
   a. Definitions
      i. "A" is called an upper bound of the sequence $\{a_n\}$ if $a_n \leq A$ for all positive integers n.
      ii. "B" is called a lower bound of the sequence $\{a_n\}$ if $a_n \geq B$ for all positive integers n.
      iii. A sequence $\{a_n\}$ is said to be bounded if there is a number "C" such that $|a_n| \leq C$ for all postive integers n. (Equivalently, $\{a_n\}$ is bounded if and only if it has an upper bound and a lower bound.)
   b. Axiom of completeness: Every set of real numbers which has a lower bound has a greatest lower bound (*glb*), and every set of real numbers which has an upper bound has a least upper bound (*lub*).
   c. Thm: A monotonic sequence is convergent if and only if it is bounded. (Note: For all sequences $\{a_n\}$ convergence implies boundedness, but not conversely.)
   d. More Theorems
      i. If $\{a_n\}$ is a monotonically increasing sequence which has an upper bound, then the sequence is convergent, its limit being the *lub* of the numbers $\{a_n\}$.

ii. If $\{a_n\}$ is a monotonically decreasing sequence which has a lower bound then the sequence is convergent, its limit being the *glb* of the numbers $\{a_n\}$.

Infinite Series

1. Definitions and Basic Theorems

a. Def'n: If $\{a_n\}$ is a sequence, then an expression of the form $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \ldots + a_m + \ldots$ is called an infinite series.

b. Def'n: Let $\sum_{n=1}^{\infty} a_n$ be an infinite series, and let $\{S_n\}$ where $S_n = a_1 + a_2 + a_3 + \ldots + a_n$ be the sequence of the partial sums of the series. If $\lim_{n\to\infty} S_n$ exists and is equal to S, the series is said to be convergent and S is called the sum of the infinite series $\sum_{n=1}^{\infty} a_n$. If $\lim_{n\to\infty} S_n$ fails to exist, the series is divergent and has no sum (If series $\sum_{n=1}^{\infty} a_n$ has a sum, we write $S = \sum_{n=1}^{\infty} a_n$).

c. Thm: (necessary condition for convergence) If the infinite series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n\to\infty} a_n = 0$. (Note: converse is not true.)

d. Corollary: If the *n*th term of an infinite series $\sum_{n=1}^{\infty} a_n$ does not approach 0 as $n \to \infty$, the series is divergent.

e. Thm: The two infinite series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=k}^{\infty} a_n$, for any positive integral k, both either converge or diverge.

f. Thm: If "c" is any non-zero constant and $\sum a_n$ converges, then so does $\sum ca_n$, and its limit is $c\sum a_n$. If $\sum a_n$ diverges, then so does $\sum ca_n$.

g. Thm: If $\sum a_n$ and $\sum b_n$ are convergent series, then $\sum (a_n + b_n)$ and $\sum (a_n - b_n)$ are also convergent, and $\sum (a_n \pm b_n) = \sum a_n \pm \sum b_n$.

h. Thm: The terms of a convergent series may be grouped in any way and the new series will converge to the same limit as the original series.

i. Thm: let $\{S_n\}$ be the sequence of partial sums of $\sum a_n$, then for every $\varepsilon > 0$ there exists a number N such that $|S_n - S_m| < \varepsilon$ whenever $n, m > N$.

2. Some Important Series

a. The Geometric Series: $\sum r^n$ is convergent for $|r| < 1$ with sum $= a/(1 - r)$ and divergent for all other values of r.

b. The Harmonic Series: $\sum 1/n$ is divergent.

c. The Hyperharmonic series $\Sigma 1/(n^k)$ is convergent for $k > 1$, and divergent for $k \leq 1$.
d. The Telescopic Series $\Sigma\ 1/(n(n+1))$ is convergent.
e. The series $\Sigma\ 1/n!$ is convergent.
f. Polynomial Test: If $P(n)$ and $Q(n)$ are polynomials of pth and qth degree respectively, then Σ $P(n)/Q(n)$ converges if $q > p + 1$ and diverges if $q \leq p + 1$.

3. Tests For Convergence of Series of Positive Term
   a. Def'n: Series of positive terms only will be called positive series.
   b. Thm: A necessary and sufficient condition for an infinite positive series to be convergent is that its sequence of partial sums has an upper bound.
   c. Thm: (Comparison Test) Let Σa_n, Σb_n, Σc_n be positive series.
      i. If Σb_n is convergent and $a_n \leq b_n$ (for all n), then Σa_n is also convergent.
      ii. If Σc_n is divergent and $a_n \geq c_n$ (for all n), then Σa_n is also divergent.
   d. Thm: (Integral Test) If f is positive, continuous, and decreasing for $x \geq 1$ and $a_n = f(n)$, then $\sum_{n=1}^{\infty} a_n$ and $\int_1^{\infty} f(x)\,dx$ either both converge or both diverge. If the series coverges to S, then the remainder $R_N = S - S_N$ is bounded by $0 \leq R_N \leq \int_N^{\infty} f(x)\,dx$.
   e. Thm: (Ratio Test) let Σa_n be a positive series, and let $\lim_{n\to\infty} (a_n + 1)/a_n = L$.
      i. If $L < 1$, the series converges.
      ii. If $L > 1$, the series diverges.
      iii. If $L = 1$, the test fails.

4. Alternating Series and Absolute Convergence
   a. Def'n: An infinite series of the form $\sum_{k=1}^{\infty} (-1)^{k+1}(a_k) = a_1 - a_2 + a_3 - \ldots$ where $a_k > 0$ (for all k), is called an alternating series.
   b. Thm: The alternating series $a_1 - a_2 + a_3 - \ldots$ converges if $0 < a_{n+1} < a_n$ (for all n) and $\lim_{n\to\infty} a_n = 0$.
   c. Thm: If $\sum_{k=1}^{\infty} (-1)^{k+1}(a_k) = a_1 - a_2 + a_3 - \ldots$ is an alternating series such that $0 < a_{k+1} < a_k$ and $\lim_{k\to\infty} a_k = 0$, the error made by using the sum of the first k terms as an approximation for the sum of the series is less than a_{k+1}. The absolute value of the first neglected term, i.e.

$$\left|\sum_{n=1}^{\infty}(-1)^{n+1}a_n - \sum_{n=k}^{\infty}(-1)^{n+1}a_n\right| < a_{k+1}$$

   d. Def'n:
      1. The infinite series Σa_n converges absolutely if $\Sigma |a_n|$ converges.
      2. If a series is convergent but not absolutely convergent, it is said to be conditionally convergent.
   e. Thm: If a series converges absolutely, then it also converges.
   f. Thm: For any series of nonzero terms Σa_n, let $\lim_{n\to\infty} |(a_{n+1})/a_n| = \mathrm{L}$.

1. If $L < 1$, series converges absolutely.
2. If $L > 1$, series diverges.

g. Thm: If a series is absolutely convergent, its terms may be rearranged without affecting the absolute convergence of the series or its sum.

h. Thm: If $\Sigma\, a_n$ and $\Sigma\, b_n$ are absolutely convergent series with sums A and B, respectively, then the series $a_1b_1 + (a_1b_2 + a_2b_1) + (a_1b_3 + a_2b_2 + a_3b_1) + \ldots$ converges absolutely and its sum is AB.

Power Series

1. Definitions and Basic Theorems

   a. Def'n: If $\{a_n\}$ is a sequence of constants, the expression $\sum_{n=0}^{\infty} a_nx^n = a_0 + a_1x + a_2x^2 + a_3x^3\ldots$ is called a power series in x.

   b. If the power series $\Sigma\, a_nx^n$ converges for a number $x_1 \neq 0$ then it converges absolutely for all number x such that $|x| < |x_1|$.

   c. If the power series diverges for a number x_2, then it diverges for all numbers x such that $|x| > |x_2|$.

   d. Let $\Sigma\, a_nx^n$ be a given power series. Then exactly one of the following conditions holds:
      i. The series converges only when $x = 0$.
      ii. The series is absolutely convergent for all values of x.
      iii. There exist a number $r > 0$ (called the radius of convergence) such that the series is absolutely convergent for all values of x for which $|x| < r$ and is divergent for all values of x for which $|x| > r$ (converges possibly at one or both endpoints r, $-r$).

   e. Def'n: The set of all values for which a power series is convergent is called the interval of convergence of the power series.

   f. Similarly, if a_n is a sequence we define the expression $\sum_{n=0}^{\infty} a_n(x - b)^n = a_0 + a_1(x - b) + a_2(x - b)^2 + a_3(x - b)^3 + \ldots$ to be a power series in $(x - b)$.

   g. The conditions in the above theorem now become:
      i. The series converges only when $x = b$.
      ii. The series is absolutely convergent for all values of x.
      iii. There exists a number $r > 0$ such that the series is absolutely convergent for all values of x for which $|x - b| < r$ and is divergent for all values of x for which $|x - b| > r$ (converges possibly at one or both endpoints $b - r$, $b + r$).

2. More on Power Series

   a. Def'n: Since the power series $\Sigma\, a_n(x - b)^n$ has a unique sum at each point in its interval of convergence, a function f is defined by the power series, i.e. $f(x) = \Sigma\, a_n(x - b)^n$. The domain of f is the interval of convergence of the power series, and the values of f at any point x in this domain is the sum of the power series at this point.

   b. Def'n: The derived series of the power series $\Sigma\, a_n(x - b)^n$ is $\Sigma\, na_n(x - b)^{n-1} = a_1 + 2a_2(x - b) + 3a_3(x - b)^2 + \ldots + na_n(x - b)^{n-1} + \ldots$, i.e., the series formed by taking the derivates of the terms of the original series.

   c. Functions defined by power series behave very much like polynomials as is indicated by the following theorem. Thm: Let $\Sigma\, a_n(x - b)^n$ be a power series in $(x - b)$ and let f be defined by $f(x) = \Sigma\, a_n(x - b)^n$, where x is any number in the interval of convergence of the series.
      i. f is continuous at every point in the interval of convergence.

ii. The derived series of the power series is equal to the derivative of the function defined by the power series at every point in the inteval of convergence, except possibly at the endpoints, i.e. $f'(x) = \sum D_x[a_n(x - b)^n] = \sum na_n(x - b)^{n-1}$.

iii. The series obtained by integrating the terms of the given series converge to $\int_b^x f(t)\,dt$ for each x within the interval of convergence of the original power series, except possibly at the endpoints, i.e. $\int_b^x f(t)\,dt = \sum_{n=0}^{\infty}\int_b^x a_n(t-b)^n dt = \sum_{n=0}^{\infty} a_n \frac{(x-b)^{n+1}}{n+1} = a_0(x-b) + a_1(x-b)^2 + \cdots$

(Note: It follows from ii above that a function that can be represented by a power series is infinitely differentiable and all derived series have the same radius of convergence (not necessarily same interval).)

3. Taylor Series and Taylor's Theorem
   a. If a function f is infinitely differentiable, the series $\sum_{n=0}^{\infty}\frac{f^{(n)}(b)(x-b)^n}{n!} = f(b) + f'(b)(x-b) + \frac{f''(b)(x-b)^2}{2!} + \frac{f'''(b)(x-b)^3}{3!} + \cdots$ is called the Taylor series in $(x - b)$ of the function.
   b. Consider any power series in $(x - b)$, $a_0 + a_1(x - b) + a_2(x - b)^2 + \ldots$ and denote its interval of convergence by $(b - r, b + r)$, where $0 \le r < \infty$. Denote by the function defined by $f(x) = a_0 + a_1(x - b) + a_2(x - b)^2 + \ldots + a_n(x - b)^n + \ldots$ where x is any number in $(b - r, b + r)$. Then the power series is the Taylor's series in $(x - b)$ of the function f, namely $f(x) = f(b) + f'(b)(x - b) + f''(b)(x - b)^2 / 2! + \ldots + f^{(n)}(b)(x - b)^n / n! + \ldots$ where $b - r < x < b + r$ (i.e. $a_n = f^{(n)}(b) / n!$).
   c. The Taylor series in x is called a Maclaurin series, $f(0) + f'(0)x + f''(0)x^2 / 2! + f'''(0)x^3 / 3! + \ldots + f^{(n)}(0)x^n / n! + \ldots$ A Taylor series in $(x - b)$ of a function f is sometimes called the Taylor expansion of f about the point b.
   d. Taylor's Theorem: Let f be a function which is defined in some open interval $(b - r, b + r)$, $0 \le r < \infty$, and whose first $n + 1$ derivatives exist and are continuous throughout the interval. Then for all numbers x in $(b - r, b + r)$, $f(x) = f(b) + f'(b)(x - b) + f''(b)(x - b)^2 / 2! + \ldots + f^{(n)}(b)(x - b)^n / n! + R_n(x)$ where $R_n(x) = f^{(n-1)}(\xi)(x - b)^{n-1} / (n + 1)!$ (Lagrange form of the remainder) where $b < \xi < x$.
   e. Other forms of the remainder are:

      i. Integral form: $R_n(x) = 1/n! \int_b^x (x - t)^n f^{(n-1)}(t)\,dt$.

      ii. Cauchy's form: $R_n(x) = f^{(n+1)}(\xi)(x - \xi)^n(x - b) / n!$, where ξ is some number between b and x.

   f. Thm: Since $f(x) - \sum_{k=0}^{n} f^{(k)}(b)(x - b)^k / k! = R_n(x)$, it follows that if f is a function having derivatives of all orders in some interval $(b - r, b + r)$. A necessary and sufficient condition that the Taylor's series $\sum_{n=0}^{\infty} f^{(n)}(b)(x - b)^n / n!$ represents the function f at all points in the interval is that, whenever $|x - b| < r$, $\lim_{n\to\infty} R_n(x) = 0$, where $R_n(x)$ is the remainder after $n + 1$ terms in Taylor's formula.

Important Series Expansions

1. $\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \cdots + \dfrac{(-1)^{n-1}x^{2n-1}}{(2n-1)!} + \cdots\ -\infty < x < \infty.$

2. $\sin x = \sin a + (x-a)\cos a - \dfrac{(x-a)^2\cos a}{2!} - \dfrac{(x-a)^3\cos a}{3!} + \cdots\ -\infty < x < \infty.$

3. $\cos x = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \cdots + \dfrac{(-1)^{n-1}x^{2n-2}}{(2n-2)!} + \cdots\ -\infty < x < \infty.$

4. $\cos x = \cos a - (x-a)\sin a - \dfrac{(x-a)^2\cos a}{2!} + \dfrac{(x-a)^3\sin a}{3!} + \cdots\ -\infty < x < \infty.$

5. $e^x = 1 + x + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \cdots + \dfrac{x^n}{n!} + \cdots\ -\infty < x < \infty.$

6. $\dfrac{1}{1-x} = 1 - x + x^2 - x^3 + \cdots\ -1 < x < 1.$

7. $\ln(1+x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \dfrac{x^4}{4} + \cdots\ -1 \le x < 1.$

8. $\dfrac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots\ -1 < x < 1.$

9. $\ln(1-x) = x + \dfrac{x^2}{2} + \dfrac{x^3}{3} + \dfrac{x^4}{4} + \cdots\ -1 < x < 1.$

10. $\ln(x) = (x-1) - \dfrac{(x-1)^2}{2} + \dfrac{(x-1)^3}{3} - \dfrac{(x-1)^4}{4} + \cdots\ 0 < x \le 2.$

11. $\dfrac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \cdots\ -1 < x < 1.$

12. $\tan^{-1} x = x - \dfrac{x^3}{3} + \dfrac{x^5}{5} - \dfrac{x^7}{7} + \cdots\ -1 \le x < 1.$

13. The General Binomial Theorem: (for all real m, $-1 < x < 1$)

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \cdots + \frac{m(m-1)\cdots(m-n+1)}{n!}x^n + \cdots$$

Does the English Language Endanger Our Concepts of Numbers?

BY LINDA LAU

Are Asian students better at taking math tests than American students? There has been much speculation in response. Irene Miura and her colleagues at San Jose State University conducted a study to see if the students' native languages was related to their mathematical understanding.

Miura felt that Asian students have "superior" skills in abstract counting, in understanding Base 10 concepts, and general achievements in math. They said that these abilities are "already apparent by first grade, before teaching effectiveness and other school-related factors can account for such large variations." One remarkable fact is that Asian languages are different from European languages in expressing numbers. Asian languages use words that directly indicate their place value, whereas European languages. In these languages, the words for 10, 11, 12, 20, and other numbers like these have no obvious relation. Ten is not related to ten, eleven, twelve, or twenty; *dix,* is not related to *onze, douze,* or *vingt* (French); *tio* is not related to *elva, tolv,* or *tjugo* (Swedish). In English and Swedish, larger numbers are more logical, but not in French. In French, *soixante-dix* (sixty-ten) means seventy, and *quatre-vingt* (four-twenty) means eighty. In contrast, in Japanese, 1 is *ichi,* 10 is *juu,* and eleven is *juu-ichi* (ten-one); twenty-one is *ni-juu-ichi* (two-ten-one). Chinese and Korean have a similar way to express numbers. "The relationship of all numerals to those from 1 to 10 and their position in the base-10 system is always seen in the words used for them."

For their experiments, they gave tests to students in America, France, Sweden, Korea, and Japan during the first half of the school year. The American students were tested at the end of the year because they weren't able to do them yet. These tests involved "their cognitive representation and their place-value understanding." None of the American first-graders tested were bilingual.

On the first test, the kids were told to read the number on the card that was shown to them out loud, and to construct it using numbered blocks. In the second test, they had to point to the ones-place and tens-place of the numbers on other cards that were shown to them. Then they saw some number blocks and they had to say what number the blocks made. Another test had them make the same number out of blocks a different way, and results showed that the American students were less able to do this than the French or Swedish, who in turn were less capable than the Korean or Japanese. Other results showed that Japanese and Korean students are also better at understanding place values. On the first question, the American first-graders had 33% correct, the French had 39%, the Swedish had 9%, the Japanese had 67%, and the Koreans had 100%. On the fourth, the percent correct were 25%, 9%, 17%, 46%, and 92% respectively. On the fifth problem, the Koreans scored 58%, the highest among the countries.

During the testing, it was observed that the Asian students chose to represent numbers with blocks representing tens and ones while the American and European ones preferred the unit blocks. The scientists commented that American and European children "comprehend the quantity represented by a whole numeral as a number of individual units, [but they] do not readily see tens and ones in written numbers, and may lack an understanding of the meaning of the individual digits as tens and ones." Miura and her colleagues attempted to do experiments with kindergartners, but the American children simply couldn't do the tasks while the Korean children could.

Miura suggested that if we want to be first in the world in math, "we need to completely restructure the first five years of life in this country, creating a radical, break-the-mold childhood."

Great Chinese Contributions to the World of Mathematics

BY SHU YING XUE

The Chinese civilization is thousands of years old and has made many valuable contributions to the modern world. Chinese mathematical discoveries, often enough, were later rediscovered by other mathematicians. The Chinese have been credited with the discovery of magic squares, the concept of the number zero, and the areas of many geometrical figures, like circles, cones, and the square pyramid.

The most influential of all the Chinese books was perhaps the *Chui-Chang Suan-Shu* or the *Nine Chapters*. This book includes 246 problems on surveying, agriculture, partnership, engineering, taxation, calculation, the solution of equations, and the properties of the right triangle. The Chinese derived correct formulae for the areas of the circle, triangle, rectangle, and trapezoids. The area of the circle was found by taking three fourths the square on the diameter or one-twelfth the square of the circumference -- a correct result if the value three was adopted for pi. However, for the area of a segment of a circle the *Nine Chapters* uses the approximate results $s(s+c)/2$, where s is the sagitta (that is, the radius minus the apothem) and c the chord or the base of the segment.

Chapter eight of the *Nine Chapters* is significant for its solution of problems in simultaneous linear equations, using both positive and negative numbers. The last problem in the chapter involves four equations with five unknowns. The ninth chapter includes problems on right - angled triangles.

In the third century, Liu Hui, an important analyst of the *Nine Chapters*, derived the figure 3.14 for pi by the use of a regular polygon of 96 sides and the approximation 3.14159 by considering a polygon of 3072 sides. Liu Hui also found the correct volume of a square pyramid. For the frustum of a circular cone a similar formula is applied, but with a value of pi as 3.

One of the most distinguished mathematicians of the Sung period (12th - 14th century) was Chu Shih - Chieh, who spent 20 years of his life as a wandering scholar, earning his living by teaching mathematics. He wrote several books and one of the most important of these was the *Precious Mirror of the Four Elements*. It dealt mainly with algebra, and includes a numerical method of solving polynomial equations which is equivalent to the 19th century "Horner's method" of the West. Thus to solve the equation $x^3 + 252x - 5292 = 0$, Chu first establishes that there is a root between 19 and 10. He then uses the transformation $y = x - 19$ to obtain the equation $y^3 + 290y - 143 = 0$, with a root between 0 and 1. His final approximate solution is $x = 19 + 14/(1 + 290)$. Similarly, to solve the equation $x^3 - 574 = 0$, he sets $y = x - 8$ to give $y^3 + 24y^2 + 192y - 62=0$, yielding as an approximate root $x = 8 + 62/(1 + 24 + 192) = 8 + 2/7$. Indeed, "Horner's method" was used by several Sung mathematicians for the numerical solution of cubic and even quartic equations.

In the fifth century AD, further advances were made in the calculation of pi. The mathematicians Tsu Ch'un-Chih and his son, Tsu Keng-Chih found an "accurate" value of pi to ten decimal places. Their means of calculating this was lost through the centuries but the discovery of the value of 3.1415929203 for pi was amazing. Nine hundred years later, the mathematician Chao Yu-Ch'in inscribed polygons in a circle with the enormous number of 16,348 sides. By doing this he confirmed the value of pi given by the Tsu family. This knowledge came to China long before similar European advances, where the value of pi was known only to the seventh decimal place over 1200 years later, by Adriaen Anthoniszoon and his son.

Liu Ju-Hsieh, a great Chinese mathematician worked on several summations of series and what bacome known as Pascal's triangle. "Pascal's" Triangle was not invented by Blaise Pascal in

1654, but rather by the Chinese in the *Precious Mirror of the Four Elements*, published in 1303. The inventor of this triangle was indeed Liu Ju-Hsieh, 427 years before the appearance of the "Pascal's" Triangle in Europe. 1100, the mathematician Haien, called it "the tabulation system for unlocking binomial coefficients."

If you slant the triangle a little to the left then you will get the "Chinese Triangle." The diagonals of "Pascal's triangle" have become the columns of the Chinese triangle. Column r = 2 contains the triangular numbers for two dimensional space; 1, 3, 6, ... The column r = 3 contains the triangular numbers for three-dimensional space; 1, 4, 10, ... Another feature of the Chinese triangle was that it contained the Fibonacci series. It will be seen if the diagonal numbers are added (whether the diagonals slope upwards to the right or downwards to the right) the sums produce the Fibonacci sequence (where a term is equal is to the sum of the two preceeding terms). It is possible that Leonardo Fibonacci may have stumbled on this series through examinating the Chinese triangle. The Fibonacci sequence; 1, 1, 2, 3, 5, 8, 13, 21, ... has been found to have many beautiful and significant properties. It contains the golden ratio (5 - 1)/2 and is found in nature.

The decimal system originated in China. Its use can be traced back to the fourteenth century BC, the archaic period known as the Shang Dynasty, though it was evidently used long before that. An example of this is an inscription from the thirteenth century BC, in which "547 days" is written "Five hundreds plus four decades plus seven days."

The fact that the decimal system existed from the very beginnings of mathematics in China gave the Chinese a substantial advantage lacking in the West, laying a foundation for most of the advances they later made. The first evidence of the proper use of decimals in Europe is found in a Spanish manuscript of 976 AD, approximately 2300 years later than the earliest Chinese evidence.

The numeral zero is believed to have been invented by Chinese mathematicians in the fourth century BC. Zero is essential to the efficient execution of mathematical computations. We know for certain that the Chinese invented the use of a blank space for the digit zero. However, the actual symbol "0," is suspected to have been invented by the Chinese but by rather another ancient civilization. The blank space on the Chinese counting board, representing zero, dates as far back as the fourth century BC. The number 405, for instance, would be "written": "four blank five" or "four hundreds, no tens, five units."

The origin of the symbol "0" for zero, as traditionally told in the West, is that it was invented in India in the ninth century AD, first seen in the inscription at Gwalior dated 870 AD. However, in actuality the zero can be traced back earlier than this. It can be seen in inscriptions in Cambodia and Sumatra, both dated 683 AD, and on Bangka Island, off Sumatra, dated 686 AD. Experts believe that these inscriptions indicated that the zero came to India from China by the way of Indochina. Absolute priority for the zero symbol is not claimed for China, since its first appearance in print was not until 1247, though there are firm grounds for believing it was used at least a century earlier. No one is for sure who invented it or when.

Both zeroes and negative terms can be found in this small section from Chu Shih-Cheih's book on algebra, *Precious Mirror of the Four Elements*, published in 1303. Each box, consisting of a group of squares containing signs, represents a "matrix" form of writing an algebraic expression. The frequent occurrence of the sign "0" for zero may be clearly seen. (In these cases it means that terms corresponding for those squares do not occur in the equation.) The diagonal lines slashed through some of the numbers in the squares indicate that they are negative numbers. (The number "one" is one vertical line, the number "two" is two vertical lines, etc.)

The Chinese also stated the existance of negative numbers. It first appeared in the *Introduction to Mathematical Studies* of 1299. The Chinese found no difficulty in understanding the concept of negative numbers. In this regard, they were seventeen hundred years ahead of the West. It was not until 630 AD that the mathematician Brahmagupta began to use negative numbers in India. In Europe, negative numbers first appeared in a book by the Greek mathematician Diophantus (about 275 AD), only to be dismissed as "absurd" when occurring as a solution of an equation. The first acceptance of negative numbers in Europe was in the mid-sixteenth century when the great Renaissance genius, Jerome Cardan (Girolamo Cardano), published his book on algebra entitled *Ars Magna*, in 1545.

The Chinese have made many great contributions to the modern world. Their innovations include the decimal system, negative numbers, the value of pi, and magic squares. Their contributions have made a strong impact on today's society. They also created the aspect of zero and the "Pascal's triangle" and the Chinese Triangle. Little wonder that the Chinese civilization has been listed as one of the greatest contributor to the world of mathematics.

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SPECIAL FEATURE

Math Team Luncheon at Bouley a Huge Success

BY IRIS LAN

The Stuyvesant High School Math Team was invited to a luncheon at *Bouley's*, the world class, five star restaurant that famous critics have hailed as "New York City's best." We had named our 6 math teams "B.O.U.L.E.Y." after the favorite restaurant of our beloved math team coach, Mr. Richard Geller. Mr. Geller showed Chef David Bouley our math team scores, and he was so impressed that he invited the whole math team to his restaurant. Representatives from all Stuyvesant math teams were there. Our principal Mr. Murray Kahn even attended and loved the food and conversation. We were served a five course meal (menu on next page); all were awestruck at the variety, colors, elegant presentation, and, most of all, taste of the food. We've never tasted anything like it. From the dried flowers pressed into our menus and elegant atmosphere to the great variety of breads, *Bouley* impressed all of us. We enjoyed our meal so much, that we have continued the math team "Bouley" name tradition: our senior math teams are named D, G, and B, the initials of Chef David G. Bouley.

We anonymously give a rating: ★★★★★

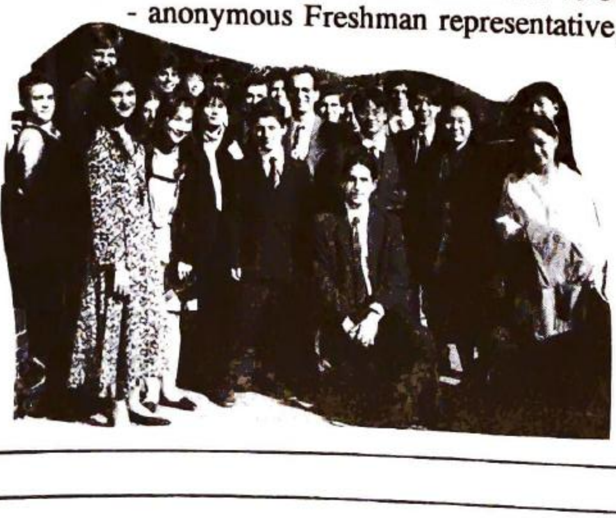
Stuyvesant math team members (acclaimed critics) say:

"The bread was scrumptious."
- Michael Develin, boy genius of B.O.U.L.E.Y.

"(Speechless.)"
- Alexander Khazanov, Gold Medal winner of I.M.O.

"The servings were just right...I was in awe."
- Oksana!

"Dessert, especially the Sorbets, was the best I've ever had in my life."
- anonymous Freshman representative, sweets expert.



Special Luncheon
for
Stuyvesant High School Math Team

Thursday October 6, 1994

...

Maine Black Sea Bass with a Stew of Roasted Sugar Pumpkin,
Blue Hubbard, White Acorn and Sweet Dumpling Squash,
with Spices and Muscovado

....

Organic Hen Roasted with Wild Mushrooms,
Puree of Chantenay Carrots from Maine, Steamed Leeks

....

Chilled Melon Soup with Fresh Fruit Sorbets

....

Hot Valrhona Chocolate Souffle with Warm Chocolate Sauce,
Vanilla and Caramel Ice Cream and Banana Tart

....

Homemade Petits Fours and Chocolates

....

Chef David Bouley

Five 5-Minute Riddles

BY IGOR TEPER

1. How many of the following ten statements are true?

1) Exactly one of these statements is false.
2) Exactly two of these statements are false.
3) Exactly three of these statements are false.
4) Exactly four of these statements are false.
5) Exactly five of these statements are false.
6) Exactly six of these statements are false.
7) Exactly seven of these statements are false.
8) Exactly eight of these statements are false.
9) Exactly nine of these statements are false.
10) Exactly ten of these statements are false.

Solution: Since all of these statements contradict each other, at most one of then can be true, which means the remaining nine are false. Therefore statement nine is the only true statement.

2. One glass contains 50 spoonfuls of wine and another glass contain 50 spoonfuls of water. A spoonful of wine is transferred to the water, and the mixture is stirred. A spoonful of the mixture is then transferred back to the glass of wine. Is there now more wine in the water, or more water in the wine?

Solution: Both glasses started out with 50 spoonfuls, and both ended up with 50 spoonfuls. This means that the amount of wine that changed glasses is equal to the amount of water that changed glasses. Therefore, the amounts are equal.

3. A man has three pheasants that he wishes to give to two fathers and two sons, giving each one pheasant. How can this be done?

Solution: He gives one to a man, another to that man's son and the third to the son of the second man.

4. How can one express 100 with four 9s?

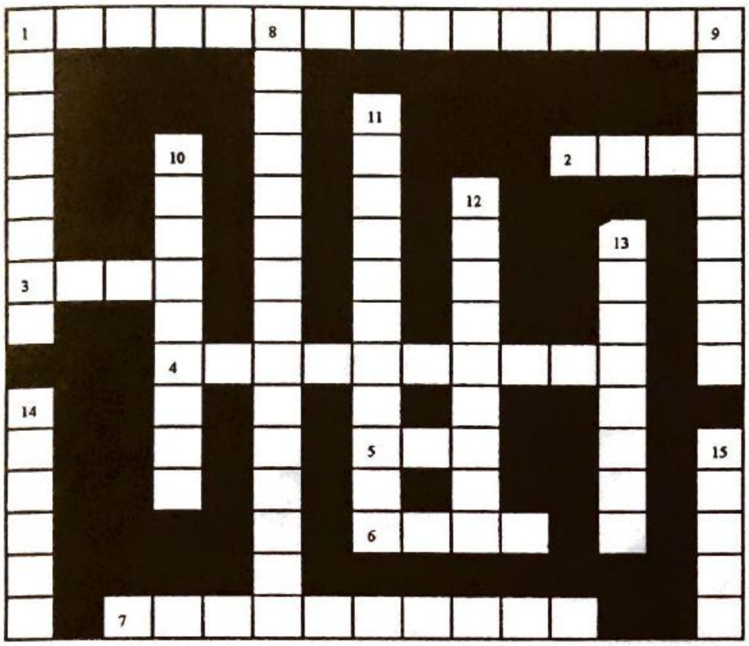
Solution: 99 9/9

5. In 1932 I was as old as the last two digits of my birth year. When I mentioned this to my grandfather, he surprised me by saying that the same applied to him. What were our ages?

Solution: The author of the problem was born in 1916, and was 16 years old. His grandfather was born in 1866 and was 66 years old.

Crossword Puzzle

BY ANTONY CHOW



Across:

1 The method of finding the slope of any curve is called __________.

2. Distance/Rate = __________.

3. 5+ 22 is a(n) __________ of $x^2 - 10x + 3$.

4. The number 2.1×10^6 is written in __________ notation.

5. Cot $\theta = 1/$__________ θ.

6. Multiplication before adding $7 \times 5 + 2$ is a(n) __________ of operation.

7. The graph of $r^2=4 \sin 2\theta$ is called a(n) __________.

Down:

1. $[r (\cos \theta + i \sin \theta)]^n = r^n (\cos n\theta + i \sin n\theta)$ is known as __________'s Theorem.

8. A method for simplifying $\sqrt{5} / \sqrt{2}$ is __________.

9. In 4/5, the number 4 is called the __________.

10. The x and y axes is called the __________ plane.

11. In 11/20, the number 20 is called the __________.

12. $1.\overline{13}$ is a(n) __________ number.

13. The x coordinate is called the __________.

14. $3x+2y=4$ is a(n) __________ equation.

15. A complex number can be expressed in __________ form.

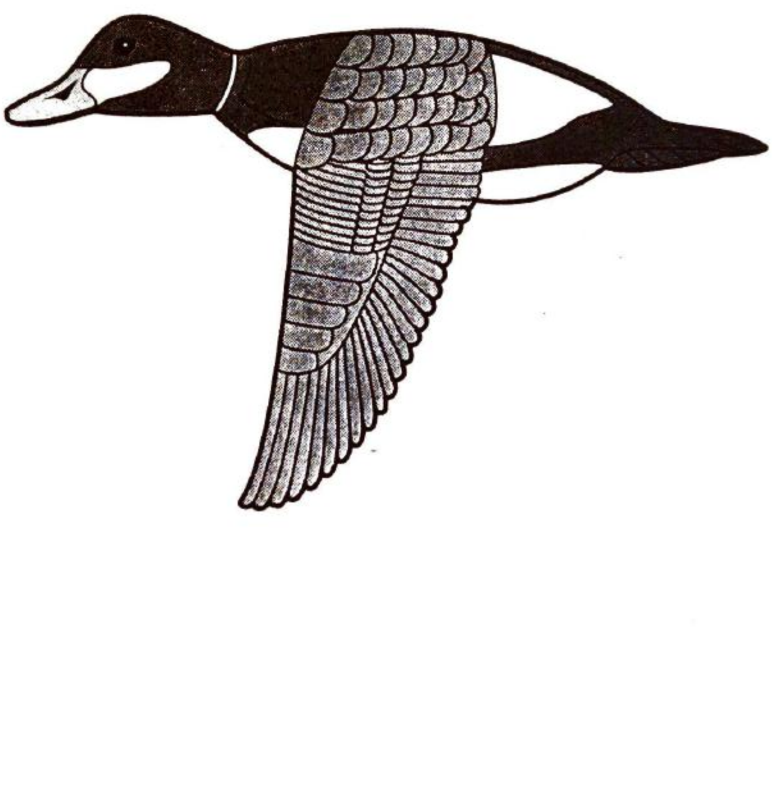
Solutions:

Across
1. Differentiation
2. Time
3. Root
4. Scientific
5. Tan
6. Rule
7. Lemniscate

Down:
1. DeMoivre
8. Rationalization
9. Numerator
10. Cartesian
11. Denominator
12. Rational
13. Abscissa
14. Linear
15. Polar

Special *Bouley* Luncheon for the Stuyvesant High School Math Team
October 6, 1994

STUYVESANT HIGH SCHOOL



New York, New York